Muhammad Ilham Rafiannandha



Senior Full Stack Engineer

Backend, payments and ISP platforms

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Professional Summary

Senior Full Stack Engineer with 6+ years building production systems where mistakes cost real money: ISP billing and provisioning, payment reconciliation, and double-entry accounting. Architected a 27-module Spring Modulith platform on 8 PostgreSQL databases that runs an ISP's full subscriber lifecycle, from FreeRADIUS/MikroTik provisioning to recurring billing and multi-gateway payment collection. Backend-first with Java/Spring Boot and Laravel, comfortable across Next.js, Vue, React Native and Flutter. Shipped 25+ products for 11 employers and clients in Indonesia, Singapore and the United States, and led a development team and internship program.

Skills

Languages:	Java, PHP, TypeScript, JavaScript, Python, Go, Dart, Kotlin, C++, SQL
Backend:	Spring Boot, Spring Modulith, Spring Security, Spring Data JPA, Laravel, NestJS, Node.js, Express, Go Fiber, REST API, JWT, OAuth
Frontend:	Next.js, React, Vue 3, Pinia, SvelteKit, Tailwind CSS, Livewire, Blade
Mobile:	React Native, Expo, Flutter, Kotlin Multiplatform
Databases:	PostgreSQL, MySQL, Redis, TimescaleDB, Firebase
DevOps & Tools:	Docker, Jenkins, GitHub Actions, CI/CD, Nginx, Linux, Git, Grafana
Payments & Fintech:	Payment Gateway Integration (Duitku, Midtrans), QRIS, Virtual Account, BRI BRIVA (host-to-host), Reconciliation, Disbursement, Double-Entry Accounting
Network / ISP:	FreeRADIUS (AAA), MikroTik RouterOS, PPPoE, RADIUS Accounting, CoA, OLT & NOC Management
Architecture:	Modular Monolith, Hexagonal Architecture, Domain-Driven Design (DDD), Event-Driven Design, Multi-Datasource Transactions, Idempotent APIs, RBAC
Spoken Languages:	Indonesian (native), English (professional working proficiency)

Work Experience

Senior Full Stack Engineer

Dec 2025 – Present

PT Surya Bestari Lestari (SBLNET), Jakarta, Indonesia | Fiber ISP for residential and business subscribers

- Architected the core ISP platform as a 27-module modular monolith (Spring Boot, Spring Modulith, hexagonal/DDD, event-driven modules) across 8 PostgreSQL databases with per-context ACID transaction management.
- Built an automated billing engine (invoicing, proration, discounts, PPN/VAT) that stores money as integer cents to prevent rounding errors, feeding a double-entry accounting module (general ledger, journal entries, chart of accounts).
- Designed a pluggable multi-gateway payment layer covering Duitku (Virtual Account, QRIS, e-wallet, retail) and a host-to-host BRI Fixed Virtual Account (BRIVA) integration with real-time inquiry and payment callbacks.
- Implemented payment settlement and reconciliation with HMAC-verified callbacks, deterministic and idempotent VA registration, and automatic invoice matching.
- Automated dunning and auto-isolation: overdue subscribers are suspended and reactivated in real time once payment lands.
- Integrated network provisioning with FreeRADIUS (AAA) and MikroTik RouterOS, including PPPoE, live session monitoring, CoA disconnect, and RADIUS accounting for data usage.
- Delivered NOC and OLT management (router health, incident tracking, device config backup), Next.js portals, and a React Native self-service app for plan upgrades, invoice payment and usage tracking.

Tech: Java, Spring Boot, Spring Modulith, Spring Security, PostgreSQL, Redis, Next.js, React Native, Jenkins, Docker

Full Stack Developer (Contract)

Mar 2025 – Dec 2025

PT Cashlez Worldwide Indonesia, Jakarta, Indonesia | Payment gateway for merchants (card, QRIS, BNPL, VA)

- Built the finance portal that verifies every settled transaction against bank and back-office records.
- Developed two-stage reconciliation: third-party settlement against bank mutation, then against back-office data.

- Automated discrepancy detection for dispute amounts and missing-in-bank or missing-in-back-office transactions.
- Built disbursement (merchant payouts) and deduction (fees, chargebacks, adjustments) modules.

Tech: Java, Spring Boot, Next.js, Jenkins, Docker

Full Stack Developer & Team Lead (Contract)
Jun 2023 – Feb 2025

PT Cyber Lab Indonesia (CoffeeLabs), Jakarta, Indonesia | IT consultancy behind the CoffeeLabs product suite

- Led the development team and ran the internship program for the CoffeeLabs product suite.
- Built an HRIS with GPS geofenced check-in and a face-recognition attendance service (Python, Node.js) to stop buddy punching.
- Delivered Point of Sale, inventory and income journal systems, and a NestJS WhatsApp gateway for automated notifications.

Tech: Laravel, Java (Spring Boot), Next.js, Vue, React Native, Python, NestJS, Jenkins, Docker

Full Stack Engineer (Freelance)
Nov 2022 – May 2023

Remedi Indonesia, Jakarta, Indonesia | Personal and corporate wellbeing platform

- Owned the mobile and backend architecture; built a cross-platform React Native app on a secure Laravel REST API.
- Implemented authentication, real-time chat and third-party integrations with data-privacy safeguards.
- Defined product scope with stakeholders and shipped iterative improvements from user feedback.

Tech: React Native, Laravel, MySQL

Full Stack Developer (Contract)
Apr 2021 – Nov 2022

PT Elteyba Medina Fauzana (Elteyba Tours), Jakarta, Indonesia | Licensed Umrah & Hajj travel agency (PPIU & PIHK)

- Built the agency platform end to end: multi-role access (admin, agent, pilgrim, coordinator) with approval flows, online registration with passport and visa upload, and package selection.
- Automated invoicing with down-payment and instalment tracking, payment reminders and Midtrans payments.
- Built real-time seat quotas, visa status tracking, and manifest generation (flight grouping, hotel room-sharing, PDFs for immigration and hotels).
- Added queue-based WhatsApp notifications with email fallback, plus a mobile pilgrim portal for itineraries, payments and documents.

Tech: Laravel, Livewire, Sanctum, Vue 3, React Native, MySQL, PostgreSQL, Redis, Midtrans, WhatsApp API

Full Stack Web Developer (Contract)
Feb 2020 – Apr 2021

PT Taiba Cococha Indonesia, Jakarta, Indonesia | Coconut-shell charcoal manufacturer and exporter

- Built a trilingual (English, Indonesian, Arabic) company website, taibacoco.co.id.
- Built a sales CRM: lead management, deal pipeline and funnel metrics, automated follow-ups, e-signed quotations.
- Secured the CRM with 2FA, role-based access control, session logs and audit trails.

Tech: Laravel, Laravel Passport, Vue 3, Pinia, MySQL, Redis

Full Stack Web Developer (Internship)
Mar 2017 – Sep 2017

TVRI, Jakarta, Indonesia

- Maintained the front end of tvri.go.id and handled full stack development for newstvri.com.

Freelance & Independent Projects

- **ProSpark Pte Ltd, Singapore** | Jul 2024 – Aug 2024: Built and maintained a Laravel LMS for workforce training.
- **Taman Safari Bogor Ticketing (client: GlobalTix Teknologi Indonesia)** | Nov 2023 – Jan 2024: Built the ticketing site, back office and a React Native (Expo) gate-checker app with QR/barcode scanning, thermal printing and forged-ticket rejection.
- **Spectro & ArkaLearn (PT Arka Spektrum Solutindo)** | Sep 2023 – Dec 2023: Built a multi-language Laravel job portal placing Indonesian workers in Japan: faceted search, application tracking, messaging and SSW visa documents.
- **Nghbr Inc, New York** | Jun 2023 – Aug 2023: Built the React Native (Expo) app for a hyper-local tech networking platform.
- **Itoya Self Photo Studio** (itoya.my.id): Booking platform (Laravel, Vue, Flutter) plus remote management for studio PCs: C++ agent with DXGI capture and H.264 streaming, uWebSockets relay, Flutter dashboard.
- **DPUPR Musi Banyuasin** (dpupr.mubakab.go.id): Government agency portal on a purpose-built Laravel CMS with nested menu builder, media library, composable home page and role-based admin.
- **mfu-macos** (open source): Python CLI for Xiaomi's official bootloader-unlock API on macOS and Linux.

Education

Bachelor of English Literature | Universitas Darma Persada, Jakarta | GPA 3.60/4.00 — **Aug 2018 – Sep 2023**

Speaking

- **Speaker**, "Web Application in the World of Work" | HMIK Talk, Pertamina University | 2022
- **Main Speaker**, "Ngoding Bareng" (Coding Together) | Permikomnas Jakarta | 2022